

MTT-I (Set-A)

Academic Year 2024-25 (Odd Semester)

Course	:	B.Tech.	Date	:	09/11/2024
Semester/ Section	:	I semester	Time Duration	:	1:30 Hrs.
Subject & Subject Code	:	Basic Mechanical Engineering & 1FY3-07	Max. Marks	:	50

Course Outcomes	
CO1	Students will be able to describe the basic mechanical engineering principles.
CO2	Students will be able to classify power plants, I.C engine, Pump, Refrigeration and Air Conditioning system.

Q. No.	CO	Questions	Marks
PART-A: Attempt All Questions (10x2 = 20 Marks)			
1.	1	State first law of thermodynamics.	2
2.	1	Describe the use of Boiler mountings and accessories.	2
3.	1	Describe thermodynamics system and surrounding.	2
4.	1	Define 'entropy' of a system.	2
5.	1	State function of Economizer and superheater in boiler.	2
6.	2	State compression ratio with reference to I C engine.	2
7.	2	Describe principle function of pump.	2
8.	2	Explain the different processes of Otto cycle.	2
9.	2	Explain strokes in compression ignition engine.	2
10.	2	Define scavenging in a 2-stroke engine	2
PART-B: Descriptive /Analytical / Problem Solving/ Design questions Attempt Any Three Questions (3x5=15 Marks)			
1.	1	Differentiate between fire tube boiler and water tube boiler.	5
2.	1	Discuss thermodynamic properties with examples.	5
3.	1	Elaborate boiler accessories with schematic diagram.	5
4.	1	Discuss Hydroelectric power plant with layout.	5

PART-C: Descriptive /Analytical / Problem Solving/ Design questions
Attempt Any Three Questions (3x5=15 Marks)

✓ 1.	2	Discuss main components of internal combustion engine with neat sketch.	5
✓ 2.	2	Elaborate working of single acting reciprocating pump with neat sketch.	5
✓ 3.	2	Compare four stroke and two stroke internal combustion engine.	5
✓ 4.	2	Explain working of centrifugal pump with neat sketch.	5